

NAFIZ KHAN

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RESEARCH PROFILE

B.Sc. graduate in Computer Science & Engineering with a research focus on the robustness & reliability of pretrained language models under quantization & weight noise. Full-stack developer (MERN) with a strong foundation in data structures, algorithms, & system design, applying that engineering rigor to building trustworthy, reproducible AI systems. Seeking a research-focused Master's/PhD position to extend this work on model robustness, security, & dependable machine learning systems.

EDUCATION

BRAC University, School of Data Science

Jan 2022 – Feb 2026

B.Sc. in Computer Science and Engineering | GPA: 3.65 / 4.00

Relevant coursework: Data Structures & Algorithms, Artificial Intelligence, Neural Networks, Computer Networks, Software Engineering, Blockchain & Cryptocurrency.

Adamjee Cantonment College

Mar 2018 – Nov 2020

Higher Secondary Certificate (HSC) | GPA: 5.00 / 5.00

Bangladesh International School

Jan 2004 – Dec 2018

Secondary School Certificate (SSC) | GPA: 5.00 / 5.00

RESEARCH EXPERIENCE

Student Researcher

Sep 2026 – Present

Elite Research Lab LLC, New York, USA

- Co-led a four-person team developing a turn-level framework to detect unhealthy relationship formation in mental health chatbot conversations; selected for the lab's 2026 Research Training and Competition.

Undergraduate Researcher

Jan 2025 – Jan 2026

BRAC University, School of Data Science

Title: Anchor-Guided Repair: Enhancing Stability of Compromised Pretrained Language Models Against Low-Precision and Weight Noise Attacks.

- Developed an anchor-guided weight repair mechanism to stabilize compromised pretrained language models without retraining.
- Demonstrated robustness against quantization and weight-noise attacks with minimal accuracy degradation.

PUBLICATIONS

Rohan, A. M., **Khan, N.**, Tanni, T. T., Fardin, F., & Bushra, A. (2026). *Anchor-Guided Repair: A Defense Mechanism for Enhancing Stability of Compromised Pretrained Language Models Against Low-Precision and Weight Noise Attacks*. In *Proceedings of IndabaX Nigeria 2026: Building Scalable AI That Works: From Research to Deployment in Resource-Constrained Environments*. *Proceedings of Machine Learning Research*, 319, 368–381. <https://proceedings.mlr.press/v319/>

ONGOING RESEARCH

Under Review

Tanni, T. T., & **Khan, N.** *Reporter Coverage and Campaign Duplication in GHSA/OSV Malware Advisories*. [Manuscript submitted for publication]. MSR 2027.

- Showed 994 GHSA/OSV malware advisories collapse to 317 independent campaigns, and that text-derived labels depend on the reporting analyst (OR ≈ 23).
- Found a well-powered null for package naming vs. payload mechanism; labeler validated by two annotators (κ 0.78/0.66); code and data released.

PROJECTS

Multimodal-RAG-Agent-for-Clothing-Store

- Built a multimodal RAG agent for clothing-store product discovery, combining image and text understanding with retrieval-augmented generation for context-aware recommendations.
- **Tools Used:** Python, LangChain, RAG, Vector Database, LLM/VLM APIs

Multi-Agent-Debate-System-with-Langchain

- Developed a multi-agent debate system using specialized AI agents to generate, challenge, and evaluate arguments through structured multi-round reasoning.
- **Tools Used:** Python, LangChain, LLM APIs

AWARDS & HONORS

- Dean's List, BRAC University, Summer 2024 & Spring 2025
- Academic Distinction, B.Sc. in Computer Science and Engineering, BRAC University, 2026
- 2nd Runner-up, CSE471 Project Demonstration, BRAC University, 2025

PROFESSIONAL EXPERIENCE

Co-founder

August 2026 – present

TriTech IT, Dhaka, Bangladesh

- Co-founded a software company delivering full-stack, AI, and automation solutions.
- Collaborating with clients and developers to deliver scalable applications.

AI & Automation Specialist

Apr 2026 – Jun 2026

Softvance Limited, Dhaka, Bangladesh

- Designed and deployed n8n automation workflows, integrating APIs and AI-driven processes with testing and optimization.

Executive, Network & IT (Intern)

Dec 2025 – Feb 2026

Wintel Limited, Dhaka, Bangladesh

- Supported computer network infrastructure, monitoring, and troubleshooting.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C, SQL

ML & AI: PyTorch, scikit-learn, LangChain, ChromaDB, NumPy, pandas, Hugging Face

Automation & MLOps: n8n, Weights & Biases, Docker, Hugging Face Spaces, Redis

Web & Databases: React.js, Node.js, Express.js, MongoDB, MySQL, SQLite, REST APIs, Supabase

Tools: Git, GitHub, LaTeX, Postman, Antigravity, Cursor, Claude code, Copilot

REFERENCES

Dr. Muhammad Iqbal Hossain

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Associate Professor, BRAC University

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Md. Kishor Morol

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